

Multimodal Deep Learning

Solutions to Exercise Sheet 8

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Exercise 1: Warm-up: The Evidence Lower Bound

In many modern generative models, our goal is to maximize the log-likelihood of observed data, $\log p_\theta(\mathbf{x})$, which is often intractable. Variational Inference (VI) reformulates this problem by maximizing a tractable, parameterized lower bound instead. This exercise will guide you through the derivation of this bound (the Evidence Lower Bound (ELBO)) and reveal its relationship with the KL divergence.

We consider a generative model with parameters θ that defines a distribution over observed data $\mathbf{x}$ and latent variables $\mathbf{z}$. The model assumes the latents generate the data, represented as $\mathbf{z} \rightarrow \mathbf{x}$.

- **Model Joint Distribution:** $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})$.
- **True Posterior:** $p_\theta(\mathbf{z}|\mathbf{x}) = \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{x})}$. This is the "true" but typically intractable distribution of the latents given the data.
- **Variational Posterior:** $q_\phi(\mathbf{z}|\mathbf{x})$. This is our tractable, parameterized approximation to the true posterior, with parameters ϕ . In the context of VAEs, this is often called the *encoder*.
- **KL Divergence:** $D_{\text{KL}}(Q\|P) = \mathbb{E}_{z \sim Q} \left[\log \frac{Q(\mathbf{z})}{P(\mathbf{z})} \right]$. A non-negative measure of dissimilarity, where $D_{\text{KL}}(Q\|P) \geq 0$.

- (a) **From KL Divergence to the ELBO** Start with the KL divergence of the variational posterior $q_\phi(\mathbf{z}|\mathbf{x})$ to the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$. Your task is to show that the log-likelihood of the data can be decomposed as follows:

$$\log p_\theta(\mathbf{x}) = D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}|\mathbf{x})) + \mathcal{L}(\theta, \phi)$$

where the Evidence Lower Bound, $\mathcal{L}(\theta, \phi)$, is defined as:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$$

Hint: Expand the KL divergence term. Since $\log p_\theta(\mathbf{x})$ does not depend on $\mathbf{z}$, it can be moved outside the expectation over q_ϕ .

- (b) **Why Maximize the ELBO?** Using the equation derived in Part A, explain why maximizing the ELBO, $\mathcal{L}(\theta, \phi)$, with respect to both model parameters θ and parameters ϕ is a valid objective. Then, state the condition under which the ELBO becomes tight (i.e., when does $\mathcal{L}(\theta, \phi) = \log p_\theta(\mathbf{x})$?).
- (c) **The Reconstruction + Regularization Form** The ELBO is often optimized in a more intuitive form. Using the product rule on the joint distribution, $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})$, show that $\mathcal{L}(\theta, \phi)$ can be rewritten as:

$$\mathcal{L}(\theta, \phi) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Term}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}))}_{\text{Regularization Term}}$$

This final form reveals the objective we actually optimize in models like Variational Autoencoders (VAEs), balancing data reconstruction with keeping the approximate posterior close to the prior.

Solution

- (a) We begin with the definition of the KL divergence between the variational posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$:

$$\begin{aligned} D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}|\mathbf{x})) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})} \right] \\ &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log q_\phi(\mathbf{z}|\mathbf{x}) - \log p_\theta(\mathbf{z}|\mathbf{x})] \end{aligned}$$

Now, we substitute the definition of the true posterior, $p_\theta(\mathbf{z}|\mathbf{x}) = \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{x})}$:

$$\begin{aligned} &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[\log q_\phi(\mathbf{z}|\mathbf{x}) - \log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{x})} \right] \\ &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log q_\phi(\mathbf{z}|\mathbf{x}) - \log p_\theta(\mathbf{x}, \mathbf{z}) + \log p_\theta(\mathbf{x})] \end{aligned}$$

We can separate the expectation. Since $\log p_\theta(\mathbf{x})$ does not depend on $\mathbf{z}$, its expectation under $q_\phi(\mathbf{z}|\mathbf{x})$ is simply itself:

$$\begin{aligned} &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log q_\phi(\mathbf{z}|\mathbf{x}) - \log p_\theta(\mathbf{x}, \mathbf{z})] + \log p_\theta(\mathbf{x}) \\ &= -\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})] + \log p_\theta(\mathbf{x}) \end{aligned}$$

Recognizing the expectation term as the definition of the ELBO, $\mathcal{L}(\theta, \phi)$, we have:

$$D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}|\mathbf{x})) = -\mathcal{L}(\theta, \phi) + \log p_\theta(\mathbf{x})$$

Rearranging this equation gives the desired decomposition of the log-likelihood:

$$\log p_\theta(\mathbf{x}) = D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}|\mathbf{x})) + \mathcal{L}(\theta, \phi)$$

- (b) From the result in Part A, we have the identity $\log p_\theta(\mathbf{x}) = D_{\text{KL}}(\dots) + \mathcal{L}(\theta, \phi)$. The KL divergence is by definition non-negative, i.e., $D_{\text{KL}}(Q \parallel P) \geq 0$. This implies:

$$\log p_\theta(\mathbf{x}) \geq \mathcal{L}(\theta, \phi)$$

Thus, $\mathcal{L}(\theta, \phi)$ is a lower bound on the log evidence of the data. Maximizing ELBO makes a valid and tractable surrogate objective for the intractable log-likelihood.

The ELBO becomes tight, i.e., $\mathcal{L}(\theta, \phi) = \log p_\theta(\mathbf{x})$, when the gap between them, which is the KL divergence term, is zero. This occurs if and only if the variational posterior is exactly equal to the true posterior:

$$D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}|\mathbf{x})) = 0 \iff q_\phi(\mathbf{z}|\mathbf{x}) = p_\theta(\mathbf{z}|\mathbf{x})$$

This is the condition under which the bound is tight.

- (c) We start with the definition of the ELBO from Part A:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$$

Using the product rule for probability on the joint distribution, we have $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})$. We substitute this into the ELBO definition:

$$\begin{aligned}\mathcal{L}(\theta, \phi) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log(p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})) - \log q_\phi(\mathbf{z}|\mathbf{x})] \\ &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z}) + \log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})]\end{aligned}$$

We now split the expectation by linearity:

$$\begin{aligned}\mathcal{L}(\theta, \phi) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] + \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})] \\ &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z})}\right]\end{aligned}$$

The first term is the expected log-likelihood of the data given the latent variables, which is the **Reconstruction Term**. The second term is by definition the KL divergence between the variational posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and the prior $p_\theta(\mathbf{z})$. This is the **Regularization Term**. Thus, we arrive at the desired form:

$$\mathcal{L}(\theta, \phi) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x} | \mathbf{z})]}_{\text{Reconstruction Term}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p_\theta(\mathbf{z}))}_{\text{Regularization Term}}$$

Exercise 2: KL Divergence to Mean Squared Error: Simplifying a Generative Objective

In this exercise, you will learn to simplify a complex objective function, a common problem in machine learning. You will use this next week in the lecture when you learn about the diffusion forward and reverse process. You will begin with the Kullback-Leibler (KL) divergence between two Gaussian distributions. Through a series of algebraic manipulations and re-parameterizations, you will prove that minimizing this divergence is equivalent to minimizing a much simpler mean squared error.

Problem Setup (Givens)

For this exercise, you are given the following mathematical objects and relationships. Treat them as axioms.

1. **Constants:** For a time index $t \in \{1, \dots, T\}$:

- $\beta_t \in (0, 1)$
- $\alpha_t = 1 - \beta_t$
- $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$

2. **Vector Construction:** A vector $\mathbf{x}_t \in \mathbb{R}^D$ is constructed from a source vector $\mathbf{x}_0 \in \mathbb{R}^D$ and a standard Gaussian noise vector $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}^D)$ according to the following equation:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

3. **Target Distribution (True Posterior):** The true posterior distribution q of $\mathbf{x}_{t-1} \in \mathbb{R}^D$ conditioned on $\mathbf{x}_t$ and $\mathbf{x}_0$ is a Gaussian:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \Sigma(t))$$

where:

- Mean: $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_t - 1}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$
- Covariance: The covariance is $\Sigma(t) = \frac{(1 - \bar{\alpha}_{t-1})(1 - \alpha_t)}{1 - \bar{\alpha}_t}$.

4. **Model Distribution (Learned Reverse Process):** We define a model p_θ with learnable parameters θ that follow a Gaussian distribution, intended to approximate q :

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I}^D)$$

5. **KL Divergence Formula:** The KL divergence of two D -dimensional diagonal Gaussians, $P_1 = \mathcal{N}(\mu_1, \sigma_1^2 \mathbf{I}^D)$ and $P_2 = \mathcal{N}(\mu_2, \sigma_2^2 \mathbf{I}^D)$, is:

$$D_{\text{KL}}(P_1 \| P_2) = \frac{1}{2} \sum_{i=1}^D \left(\log \frac{\sigma_2^2}{\sigma_1^2} - 1 + \frac{\sigma_1^2}{\sigma_2^2} + \frac{(\mu_{1,i} - \mu_{2,i})^2}{\sigma_2^2} \right)$$

(a) **Simplifying the Objective via Fixed Variance**

Our goal is to optimize the model parameters θ by minimizing the KL divergence of the target distribution q to the model distribution p_θ . The objective function for a given time step t is:

$$L_t(\theta) = D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

To simplify the problem, we make a crucial modeling choice: we fix the variance of our model p_θ to be equal to the variance of the target q . That is, we set the model's variance σ_t^2 to the known value of the variance of the variational posterior, $\Sigma(t)$.

Your Task: Show that with the choice $\sigma_t^2 = \Sigma(t)$, minimizing $L_t(\theta)$ with respect to θ is equivalent to minimizing the squared Euclidean distance between the means of the two distributions:

$$\min_{\theta} L_t(\theta) \iff \min_{\theta} \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2$$

Hint: The parameters θ only influence μ_θ . Identify all terms in the KL divergence formula that do not depend on θ and can thus be treated as constants during optimization.

(b) **Re-parameterizing the Target Mean**

The expression for the target mean $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$ depends on $\mathbf{x}_0$, which is unavailable during the reverse process. We can re-express this mean entirely in terms of the known $\mathbf{x}_t$ and the noise vector ϵ that generated it.

Your Task: Start by isolating $\mathbf{x}_0$ from the $\mathbf{x}_t$ definition (Given #2). Then substitute this into the expression for $\tilde{\mu}_t$ (Given #3) and group the terms for $\mathbf{x}_t$ and ϵ . Through algebraic simplification, prove that the target mean can be rewritten as:

$$\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right)$$

Hint: This is a direct algebraic manipulation. You will need to use the identities $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$ and $\beta_t = 1 - \alpha_t$ to simplify the coefficients of the $\mathbf{x}_t$ and ϵ terms.

(c) **Deriving the Final, Practical Objective**

We now use the insight from Part B to re-parameterize our model's mean. Instead of designing a network to predict the complex mean μ_θ directly, we will design a network

$\epsilon_\theta(\mathbf{x}_t, t)$ that attempts to predict the noise vector ϵ from the noisy vector $\mathbf{x}_t$. The model's mean is then constructed using the same functional form as the true mean:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$$

Your Task: Substitute this new definition for μ_θ and the simplified expression for $\tilde{\mu}_t$ from Part B into the simplified objective from Part A. Show that this yields an objective proportional to the mean squared error between the true noise ϵ and the predicted noise ϵ_θ . Specifically, show that:

$$\|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2 = C \cdot \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2$$

where C is a positive constant that does not depend on the model parameters θ . This proves that training the model is equivalent to a simple noise-prediction task. Determine the expression for C in terms of α_t , β_t , and $\bar{\alpha}_t$.

This final result proves that minimizing the original, complex KL divergence objective is equivalent to training a neural network on a simple noise-prediction task using a standard MSE loss. After removing the constant weighting term C , we arrive at the famous L_{simple} objective from the DDPM paper <https://arxiv.org/pdf/2006.11239>

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2]$$

Solution

- (a) The objective is $L_t(\theta) = D_{\text{KL}}(q \| p_\theta)$. We are given the distributions are D -dimensional diagonal Gaussians: $q = \mathcal{N}(\tilde{\mu}, \Sigma(t) \mathbf{I}^D)$ and $p_\theta = \mathcal{N}(\mu_\theta, \sigma_t^2 \mathbf{I}^D)$. The KL divergence formula is:

$$D_{\text{KL}}(q \| p_\theta) = \frac{1}{2} \sum_{i=1}^D \left(\log \frac{\sigma_t^2}{\Sigma(t)} - 1 + \frac{\Sigma(t)}{\sigma_t^2} + \frac{(\tilde{\mu}_i - \mu_{\theta,i})^2}{\sigma_t^2} \right)$$

We make the simplifying choice to fix the model's variance to that of the target posterior: $\sigma_t^2 = \Sigma(t)$. Substituting this into the formula gives:

$$\begin{aligned} L_t(\theta) &= \frac{1}{2} \sum_{i=1}^D \left(\log \frac{\Sigma(t)}{\Sigma(t)} - 1 + \frac{\Sigma(t)}{\Sigma(t)} + \frac{(\tilde{\mu}_i - \mu_{\theta,i})^2}{\Sigma(t)} \right) \\ &= \frac{1}{2} \sum_{i=1}^D \left(\log(1) - 1 + 1 + \frac{(\tilde{\mu}_i - \mu_{\theta,i})^2}{\Sigma(t)} \right) \\ &= \frac{1}{2} \sum_{i=1}^D \left(0 + \frac{(\tilde{\mu}_i - \mu_{\theta,i})^2}{\Sigma(t)} \right) \\ &= \frac{1}{2\Sigma(t)} \sum_{i=1}^D (\tilde{\mu}_i - \mu_{\theta,i})^2 \end{aligned}$$

The sum is the definition of the squared L2-norm: $\sum_{i=1}^D (\tilde{\mu}_i - \mu_{\theta,i})^2 = \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2$. So, our objective becomes:

$$L_t(\theta) = \frac{1}{2\Sigma(t)} \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2$$

When minimizing $L_t(\theta)$ with respect to θ , the term $\frac{1}{2\Sigma(t)}$ is a positive constant. The parameters θ only affect μ_θ . Therefore, minimizing $L_t(\theta)$ is equivalent to minimizing the squared Euclidean distance between the means:

$$\min_{\theta} L_t(\theta) \iff \min_{\theta} \|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|_2^2$$

(b) We start with the definition of $\mathbf{x}_t$:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

First, we isolate $\mathbf{x}_0$:

$$\sqrt{\bar{\alpha}_t} \mathbf{x}_0 = \mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon \implies \mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon \right)$$

Next, we substitute this expression for $\mathbf{x}_0$ into the definition of $\tilde{\mu}$:

$$\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$$

Substituting $\mathbf{x}_0$:

$$\tilde{\mu} = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \left(\frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon \right) \right) + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$$

Using the identity $\sqrt{\bar{\alpha}_t} = \sqrt{\alpha_t \bar{\alpha}_{t-1}} = \sqrt{\alpha_t} \sqrt{\bar{\alpha}_{t-1}}$, we get $\frac{\sqrt{\bar{\alpha}_{t-1}}}{\sqrt{\bar{\alpha}_t}} = \frac{1}{\sqrt{\alpha_t}}$:

$$= \frac{\beta_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \left(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon \right) + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$$

Now we group the coefficients for $\mathbf{x}_t$ and ϵ :

$$= \left(\frac{\beta_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \right) \mathbf{x}_t - \left(\frac{\beta_t \sqrt{1 - \bar{\alpha}_t}}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \right) \epsilon$$

For the $\mathbf{x}_t$ coefficient, we find a common denominator $\sqrt{\alpha_t}(1 - \bar{\alpha}_t)$: $\frac{\beta_t + \alpha_t(1 - \bar{\alpha}_{t-1})}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} = \frac{(1 - \alpha_t) + \alpha_t - \alpha_t \bar{\alpha}_{t-1}}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} = \frac{1 - \bar{\alpha}_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} = \frac{1}{\sqrt{\alpha_t}}$

$$\tilde{\mu} = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon$$

Finally, factoring out $\frac{1}{\sqrt{\alpha_t}}$ gives the desired result:

$$= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right)$$

(c) From Part A, our simplified objective is to minimize $\|\tilde{\mu} - \mu_\theta\|_2^2$. We use the expression for $\tilde{\mu}$ from Part B and the given re-parameterization for μ_θ :

$$\begin{aligned} \tilde{\mu} &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) \\ \mu_\theta &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) \end{aligned}$$

Now, we compute their difference:

$$\begin{aligned} \tilde{\mu} - \mu_\theta &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta \right) \\ &= \frac{1}{\sqrt{\alpha_t}} \left[(\mathbf{x}_t - \mathbf{x}_t) - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon + \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta \right] \\ &= \frac{1}{\sqrt{\alpha_t}} \left[\frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} (\epsilon_\theta - \epsilon) \right] \\ &= \frac{\beta_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} (\epsilon_\theta - \epsilon) \end{aligned}$$

Finally, we compute the squared L2-norm of this difference:

$$\begin{aligned}\|\tilde{\mu} - \mu_\theta\|_2^2 &= \left\| \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}(\epsilon_\theta - \epsilon) \right\|_2^2 \\ &= \left(\frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}} \right)^2 \|\epsilon_\theta - \epsilon\|_2^2\end{aligned}$$

Since $\|\mathbf{a} - \mathbf{b}\|_2^2 = \|\mathbf{b} - \mathbf{a}\|_2^2$, this is equivalent to:

$$= \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2$$

This shows that the squared distance between the means is proportional to the mean squared error between the true noise ϵ and the predicted noise ϵ_θ . The constant of proportionality, C , is:

$$C = \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)}$$

Since C is a positive constant that does not depend on the learnable parameters θ , minimizing $\|\tilde{\mu} - \mu_\theta\|_2^2$ is equivalent to minimizing $\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2$.